

**BACHELOR OF COMPUTER SCIENCE
SCHOOL OF COMPUTER SCIENCE
BINA NUSANTARA UNIVERSITY
JAKARTA**

ASSESSMENT FORM

Course: SCIE6063001– Introduction to Computational Physics

Method of Assessment: Case Study

Semester/Academic Year: 3/2023–2024

Name of Lecturer : Agung Trisetyarso, S.Si., M.Si., Ph.D.

Date : January 6, 2024

Class : LB01

Topic :

- Voltage and Current**
- Resistance and Capacitance**
- Circuits Design & Analysis**
- Induction and Inductance**
- Electromagnetic Oscillation and Alternating Current**

Group Members:	1 Jonathan Alvindo Fernandi
	2 _____
	3 _____
	4 _____
	5 _____
	6 _____
	7 _____
	8 _____

Student Outcomes:

(SO 1) Mampu menganalisis masalah komputasi yang kompleks dan mengaplikasikan prinsip komputasi dan keilmuan lain yang sesuai untuk mengidentifikasi solusi.

Able to analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.

Learning Objectives:

(LObj 1.2) Mampu menerapkan prinsip komputasi dan disiplin ilmu terkait lainnya untuk mengidentifikasi solusi.

Able to apply principles of computing and other relevant disciplines to identify solutions.

No.	Assessment Criteria	Weight	Excellent (85–100)	Good (75–84)	Average (65–74)	Poor (0–64)	Score	(Score × Weight)
1	Ability to use potential method, serial, and parallel connection concept to provide the solution	50%	Able to solve in both numerical and computational approaches well without errors.	Able to solve in both numerical and computational approaches well with some errors.	Able to solve in either numerical or computational approaches well with some errors.	Able to solve in either numerical or computational approaches with major errors, or not able to solve at all.		
2	Ability to solve problems in topics of Voltage and Current, Resistance and Capacitance, Electromagnetic Oscillation and Alternating Current	50%	Able to solve in both numerical and computational approaches well without errors.	Able to solve in both numerical and computational approaches well with some errors.	Able to solve in either numerical or computational approaches well with some errors.	Able to solve in either numerical or computational approaches with major errors.		
Total Score: $\sum(\text{Score} \times \text{Weight})$								

Remarks:

ASSESSMENT METHOD

Instructions

Individual Assignment

Answer these problems using a numerical approach and a computational approach using Python Power Electronics. Besides your numerical answer, attach your circuit topology, parameters, and simulation output graph. Submit your answer in .pdf format.

Problem 1

From the following circuit, prove your manual calculation using PPE simulation to calculate the parameters below:

1. Total current flow in the circuit
2. Potential difference at each end of the resistance
3. The amount of current that passes through resistance 2 and resistance 3

Answer:

a. Numerical Approach

1. Total current flow in the circuit

$$R_1 = 2 \, \Omega$$

$$R_2 = 2 \, \Omega$$

$$R_3 = 2 \, \Omega$$

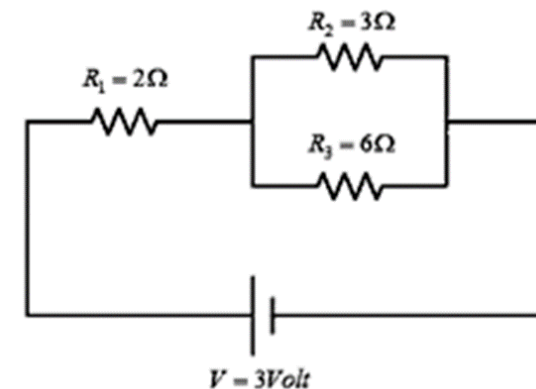
$$V = 3 \, V$$

- 1) Resistor total

- i. Resistor paralel

$$\frac{1}{R_p} = \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{3} + \frac{1}{6}$$



$$\frac{1}{R_p} = \frac{1}{2}$$

$$R_p = 2 \, \Omega$$

ii. Resistor seri

$$R_s = R_1 + R_p$$

$$R_s = 2 + 2$$

$$R_s = 4 \, \Omega$$

$$R_t = R_s$$

$$R_t = 4 \, \Omega$$

2) Total current

$$I_t = \frac{V}{R_t}$$

$$I_t = \frac{3}{4}$$

$$I_t = 0,75 \, \text{A}$$

2. Potential difference at each end of the resistance

$$I_t = 0,75 \, \text{A}$$

$$R_1 = 2 \, \Omega$$

$$R_p = 2 \, \Omega$$

$$V_1 = I_t \cdot R_1$$

$$V_1 = 0,75 \cdot 2$$

$$V_1 = 1,5 \, \text{V}$$

$$V_2 = I_t \cdot R_p$$

$$V_2 = 0,75 \cdot 2$$

$$V_2 = 1,5 \, \text{V}$$

$$V_3 = I_t \cdot R_p$$

$$V_3 = 0,75 \cdot 2$$

$$V_3 = 1,5 \text{ V}$$

3. The amount of current that passes through resistance 2 and resistance 3

$$V_2 = 1,5 \text{ V}$$

$$R_2 = 3 \, \Omega$$

$$V_3 = 1,5 \text{ V}$$

$$R_3 = 6 \, \Omega$$

$$I_2 = \frac{V_2}{R_2}$$

$$I_2 = \frac{1,5}{3}$$

$$I_2 = 0,5 \text{ A}$$

$$I_3 = \frac{V_3}{R_3}$$

$$I_3 = \frac{1,5}{6}$$

$$I_2 = 0,25 \text{ A}$$

b. Computational Approach using PPE

1) Circuit topology

[illegible]

2) Circuit parameters

1. Component type: Ammeter
Component name: load1
Component position: 6E
Positive direction of current: 6F

Ammeter positive direction of current: searah jarum jam

2. Component type: Ammeter
Component name: load2
Component position: 3L
Positive direction of current: 3M

Resistor_source value = 0.01Ω (bernilai rendah guna menjaga stabilitas tegangan dan arus)

3. Component type: Ammeter
Component name: load3
Component position: 9L
Positive direction of current: 9M

VoltageSource_dcsource peak value = $V \cdot \sqrt{2}$

VoltageSource_dcsource peak value = $3 \cdot \sqrt{2}$

VoltageSource_dcsource peak value = 4,243 V

4. Component type: Ammeter
Component name: source
Component position: 16C
Positive direction of current: 16B

Voltmeter_load voltage level = 1000 V (memerlukan tegangan tinggi)

5. Component type: Resistor
Component name: load1
Component position: 6C
Resistor value: 2.0

6. Component type: Resistor
Component name: load2
Component position: 3J
Resistor value: 3.0

7. Component type: Resistor
Component name: load3
Component position: 9J
Resistor value: 6.0

8. Component type: Resistor
Component name: source
Component position: 16E
Resistor value: 0.01

9. Component type: VoltageSource
Component name: dcsource
Component position: 16G
Peak value: 4.243
Frequency: 60.0
Phase angle: 0.0
Dc offset: 0.0
Positive polarity: 16F

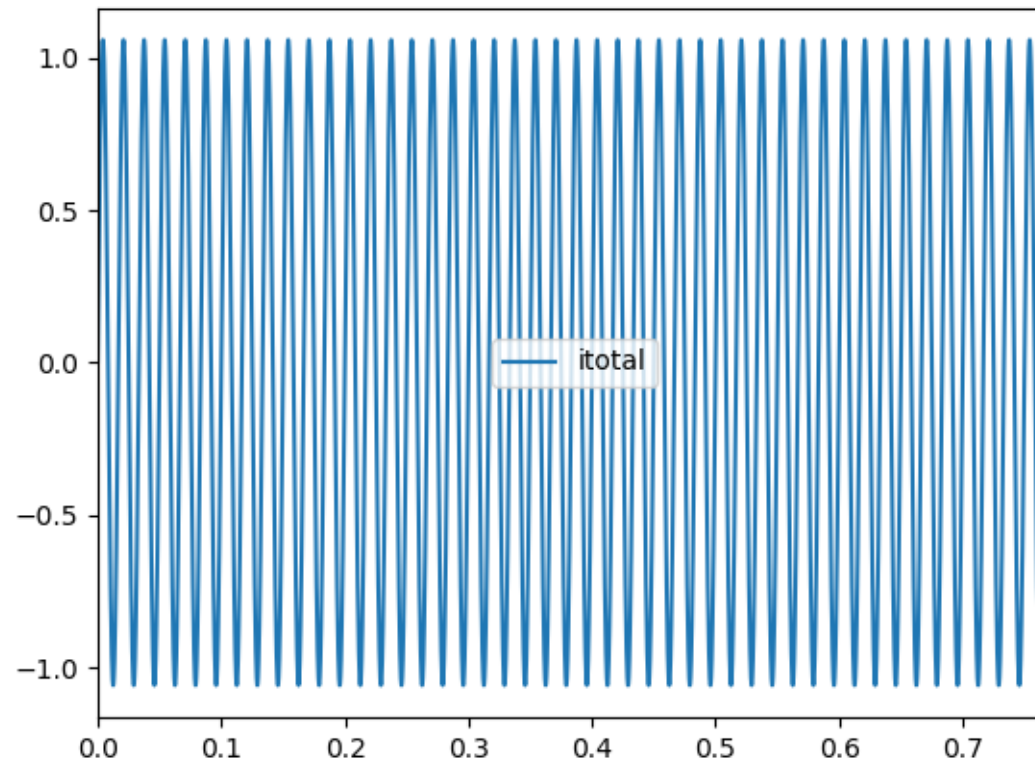
10. Component type: Voltmeter
Component name: load1
Component position: 3D
Voltage level: 1000.0
Positive direction of voltage: 3C

11. Component type: Voltmeter
Component name: load2
Component position: 1J
Voltage level: 1000.0
Positive direction of voltage: 1I

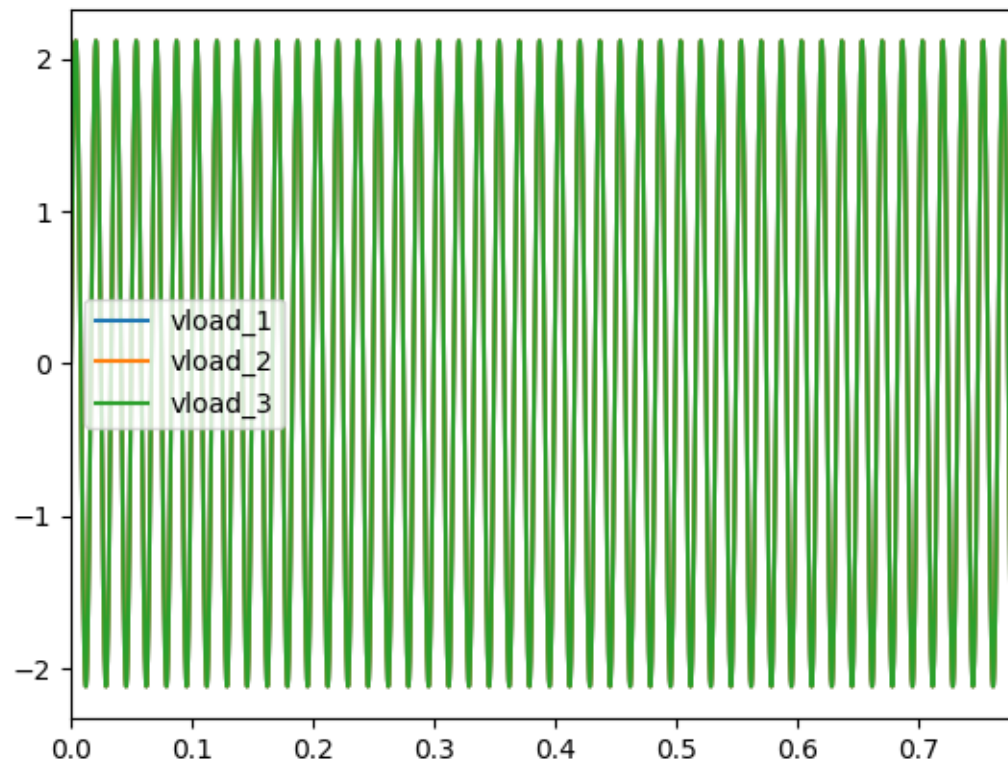
12. Component type: Voltmeter
Component name: load3
Component position: 11J
Voltage level: 1000.0
Positive direction of voltage: 11I

3) Simulation output graph

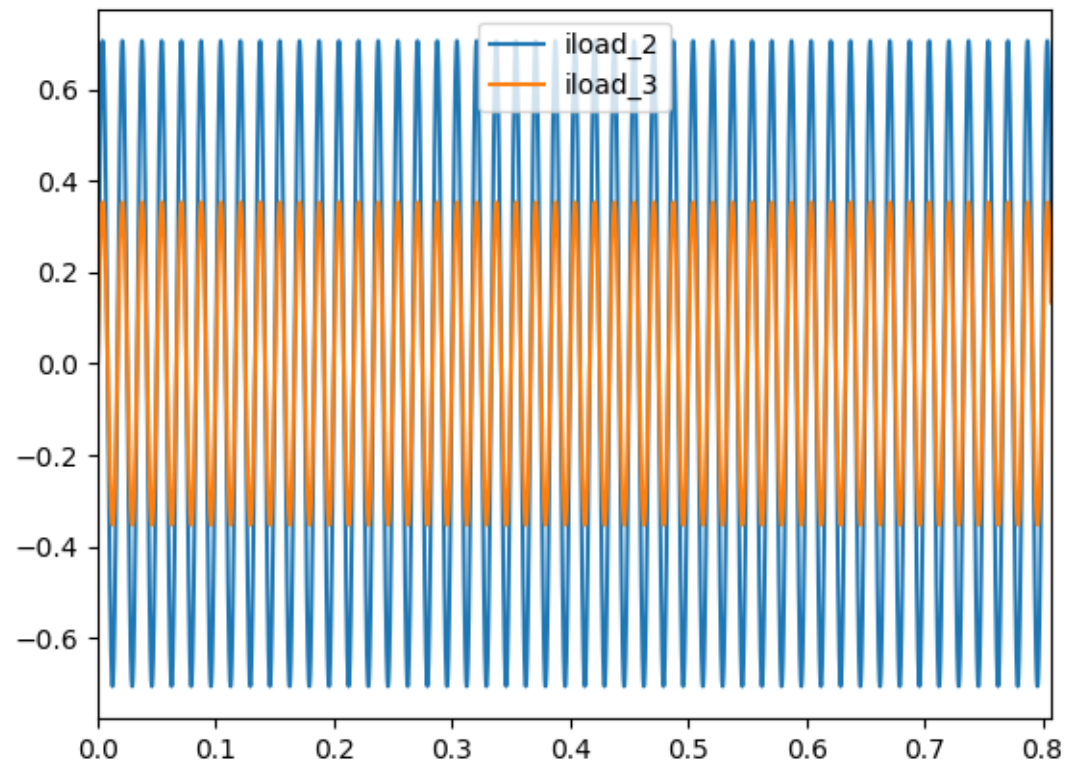
1. Total current flow in the circuit



2. Potential difference at each end of the instance



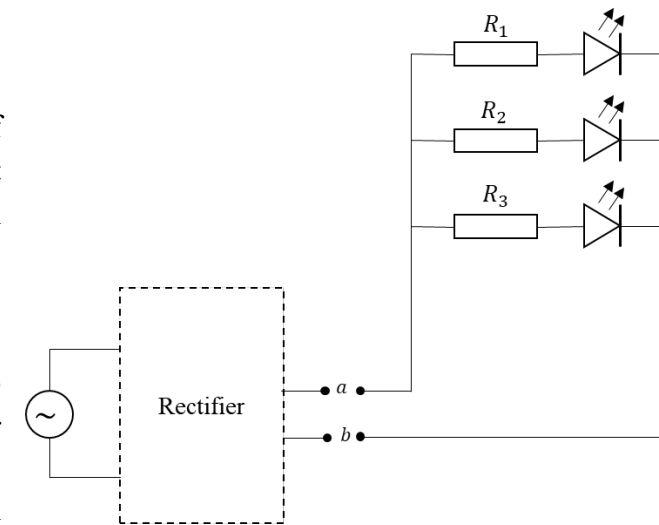
3. The amount of current that passes through resistance 2 and resistance 3



Problem 2

You are planning to conduct a small electronic project. You are planning to make a simple LED circuit (see diagram). The circuit contains three green LED lights that are connected in parallel. From the LED specification sheet, you know that the LED will work at a minimum voltage of 2 V . Also, you gain information that the LED will break if the current that flows through it exceeds 20 mA . For that, you need to use some resistors to limit the current that flows through the LED.

To power the circuit, you are planning to use a micro hydro generator that you already have. However, you realize that you need to convert the current from AC to DC so you can light up the LED. But you don't have the proper converter at the moment. So, you decide to make your own rectifier, *a basic RLC rectifier*, from only the components you have (see table) at hand. Assume that your component supply is large enough, so you can use any number of each component.



Assume that your LED does not have any internal resistance. The LED circuit will be connected to the rectifier at points *a* and *b* (see diagram). The generator has an output of 5 V with a frequency of 20 Hz . **So, what is your solution for the basic RLC rectifier?** Assume that each component in the table (including the EMF) has an internal resistance of $0.1\ \Omega$.

No	Component	Value
1	Resistor	$3\ \Omega$
2		$24\ \Omega$
3		$36\ \Omega$
4		$100\ \Omega$
5		$130\ \Omega$
6		$220\ \Omega$
7		$510\ \Omega$
8	Capacitor	$1.0\ \mu\text{F}$
9		$3.3\ \mu\text{F}$
10		$22\ \mu\text{F}$
11	Inductor	$2\ \mu\text{H}$
12		$5.1\ \mu\text{H}$
13		2 mH
14		400 mH

Answer:

a. Numerical Approach

Minimum voltage for the LED to work = 2 V

LED maximum current = 20 mA

LED maximum current = $20 \cdot 10^{-3}$ A

LED maximum current = 0,02 A

Generator output voltage = 5 V

Frequency = 20 Hz

Internal resistance = 0,1 Ω

Resistor, capacitor, and inductor values:

No	Component	Value
1	Resistor	3 Ω
2		24 Ω
3		36 Ω
4		100 Ω
5		130 Ω
6		220 Ω
7		510 Ω
8	Capacitor	1.0 μF
9		3.3 μF
10		22 μF
11	Inductor	2 μH
12		5.1 μH
13		2 mH
14		400 mH

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{1} + \frac{1}{1} + \frac{1}{1}$$

$$\frac{1}{R_p} = 3$$

$$R_p = \frac{1}{3} \Omega$$

1. Resistor value

1) 3 Ω Resistor

$$R = \frac{1}{3} \cdot 3$$

$$R = 1 \Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{2}{1}$$

$$I = 2 \text{ A (tidak dapat digunakan karena } I > 0,02 \text{ A)}$$

2) 24 Ω Resistor

$$R = \frac{1}{3} \cdot 24$$

$$R = 8 \Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{2}{8}$$

$$I = 0,25 \text{ A (tidak dapat digunakan karena } I > 0,02 \text{ A)}$$

3) 36 Ω Resistor

$$R = \frac{1}{3} \cdot 36$$

$$R = 12 \Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{2}{12}$$

$$I = 0,167 \text{ A (tidak dapat digunakan karena } I > 0,02 \text{ A)}$$

4) 100 Ω Resistor

$$R = \frac{1}{3} \cdot 100$$

$$R = 33,333 \, \Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{2}{33,333}$$

$$I = 0,06 \, \text{A} \text{ (tidak dapat digunakan karena } I > 0,02 \, \text{A)}$$

5) 130 Ω Resistor

$$R = \frac{1}{3} \cdot 130$$

$$R = 43,333 \, \Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{2}{43,333}$$

$$I = 0,046 \, \text{A} \text{ (tidak dapat digunakan karena } I > 0,02 \, \text{A)}$$

6) 220 Ω Resistor

$$R = \frac{1}{3} \cdot 220$$

$$R = 73,333 \, \Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{2}{73,333}$$

$$I = 0,027 \, \text{A} \text{ (tidak dapat digunakan karena } I > 0,02 \, \text{A)}$$

7) 510 Ω Resistor

$$R = \frac{1}{3} \cdot 510$$

$$R = 170 \, \Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{2}{170}$$

$I = 0,012 \text{ A}$ (dapat digunakan karena $I \leq 0,02 \text{ A}$)

$$R_1 = R_2 = R_3 = 510 \, \Omega$$

2. Generator resistor

1) $3 \, \Omega$ Resistor

$$I = \frac{V}{R}$$

$$I = \frac{5}{3}$$

$I = 1,667 \text{ A}$ (tidak digunakan karena I kurang mendekati $0,02 \text{ A}$)

2) $24 \, \Omega$ Resistor

$$I = \frac{V}{R}$$

$$I = \frac{5}{24}$$

$I = 0,208 \text{ A}$ (tidak digunakan karena I kurang mendekati $0,02 \text{ A}$)

3) $36 \, \Omega$ Resistor

$$I = \frac{V}{R}$$

$$I = \frac{5}{36}$$

$I = 0,139 \text{ A}$ (tidak digunakan karena I kurang mendekati $0,02 \text{ A}$)

4) $100 \, \Omega$ Resistor

$$I = \frac{V}{R}$$

$$I = \frac{5}{100}$$

$I = 0,05 \text{ A}$ (tidak digunakan karena I kurang mendekati $0,02 \text{ A}$)

5) $130 \, \Omega$ Resistor

$$I = \frac{V}{R}$$

$$I = \frac{5}{130}$$

$I = 0,038$ A (tidak digunakan karena I kurang mendekati $0,02$ A)

6) 220Ω Resistor

$$I = \frac{V}{R}$$

$$I = \frac{5}{220}$$

$I = 0,023$ A (tidak digunakan karena I kurang mendekati $0,02$ A)

7) 510Ω Resistor

$$I = \frac{V}{R}$$

$$I = \frac{5}{510}$$

$I = 0,01$ A (tidak digunakan karena I kurang mendekati $0,02$ A)

$r_{load} = 220 \Omega$

b. Computational Approach using PPE

1) Circuit topology

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
1																wire	wire	wire	wire	Voltmeter_load1	wire	wire	wire	wire	wire
2																wire			wire				wire		wire
3																wire			wire	Resistor_load1	wire	Ammeter_load1	wire		wire
4																wire									wire
5																wire	wire	wire	wire	Voltmeter_load2	wire	wire	wire	wire	wire
6																wire			wire				wire		wire
7																wire			wire	Resistor_load2	wire	Ammeter_load2	wire		wire
8																wire									wire
9																wire	wire	wire	wire	Voltmeter_load3	wire	wire	wire	wire	wire
10																wire			wire				wire		wire
11																wire			wire	Resistor_load3	wire	Ammeter_load3	wire		wire
12																wire									wire
13																wire									wire
14																wire									wire
15																wire									wire
16																wire									wire
17	wire	wire	Ammeter_Isource	wire	Resistor_Ramm	wire	Diode_Dfilter	Inductor_Ifilter	wire	Resistor_Rfilter	wire	wire	wire	wire	wire	wire									wire
18	wire										wire														wire
19	wire										wire														wire
20	wire	wire	wire								wire	wire	wire												wire
21	wire		wire								wire		wire												wire
22	wire		wire								wire		wire												wire
23	Resistor_Rsource		wire								Capacitor_Cfilter		wire												wire
24	wire		Voltmeter_Vsource								wire		Voltmeter_Cfilter												wire
25	VoltageSource_Vsource		wire								Resistor_Cfilter		wire												wire
26	wire		wire								wire		wire												wire
27	wire		wire								wire		wire												wire
28	wire	wire	wire								wire	wire	wire												wire
29	wire										wire														wire
30	wire										wire														wire
31	wire	wire	wire	wire	wire	wire	wire	Resistor_Rload	wire	wire	wire	wire	wire	wire	wire	wire	wire	wire	wire	wire	wire	wire	wire	wire	wire

2) Circuit parameters

1.	Component type: Ammeter Component name: Isource Component position: 17C Positive direction of current: 17D	13.	Component type: Resistor Component name: load1 Component position: 3T Resistor value: 510.0
2.	Component type: Ammeter Component name: load1 Component position: 3V Positive direction of current: 3W	14.	Component type: Resistor Component name: load2 Component position: 7T Resistor value: 510.0
3.	Component type: Ammeter Component name: load2 Component position: 7V Positive direction of current: 7W	15.	Component type: Resistor Component name: load3 Component position: 11T Resistor value: 510.0
4.	Component type: Ammeter Component name: load3 Component position: 11V Positive direction of current: 11W	16.	Component type: VoltageSource Component name: Vsource Component position: 25A Peak value: 7.071 Frequency: 20.0 Phase angle: 0.0 Dc offset: 0.0 Positive polarity: 24A
5.	Component type: Capacitor Component name: Cfilter Component position: 23K Capacitor value: 1e-06 Positive polarity: 24K	17.	Component type: Voltmeter Component name: Cfilter Component position: 24M Voltage level: 1000.0 Positive direction of voltage: 23M
6.	Component type: Diode Component name: Dfilter Component position: 17G Voltage level: 1000.0 Direction of cathode: 17H	18.	Component type: Voltmeter Component name: Vsource Component position: 24C Voltage level: 1000.0 Positive direction of voltage: 23C
7.	Component type: Inductor Component name: Ifilter Component position: 17H Inductor value: 0.001	19.	Component type: Voltmeter Component name: load1 Component position: 1T Voltage level: 1000.0 Positive direction of voltage: 1S
8.	Component type: Resistor Component name: Cfilter Component position: 25K Resistor value: 0.01	20.	Component type: Voltmeter Component name: load2 Component position: 5T Voltage level: 1000.0 Positive direction of voltage: 5S
9.	Component type: Resistor Component name: Ramm Component position: 17E Resistor value: 0.01	21.	Component type: Voltmeter Component name: load3 Component position: 9T Voltage level: 1000.0 Positive direction of voltage: 9S
10.	Component type: Resistor Component name: Rfilter Component position: 17J Resistor value: 0.01		
11.	Component type: Resistor Component name: Rload Component position: 31H Resistor value: 220.0		
12.	Component type: Resistor Component name: Rsource Component position: 23A Resistor value: 0.01		

Ammeter positive direction of current: searah jarum jam

Capacitor_Cfilter value = 1 uF

Capacitor_Cfilter value = $1 \cdot 10^{-6}$ F

Capacitor_Cfilter value = 1e-06 F

Diode voltage level = Voltmeter voltage level = 1000 V (memerlukan tegangan tinggi)

Diode direction of cathode: searah jarum jam

Voltmeter positive direction of voltage: berlawanan arah jarum jam

Inductor value = 1 mH

Inductor value = $1 \cdot 10^{-3}$ H

Inductor value = 0,001 H

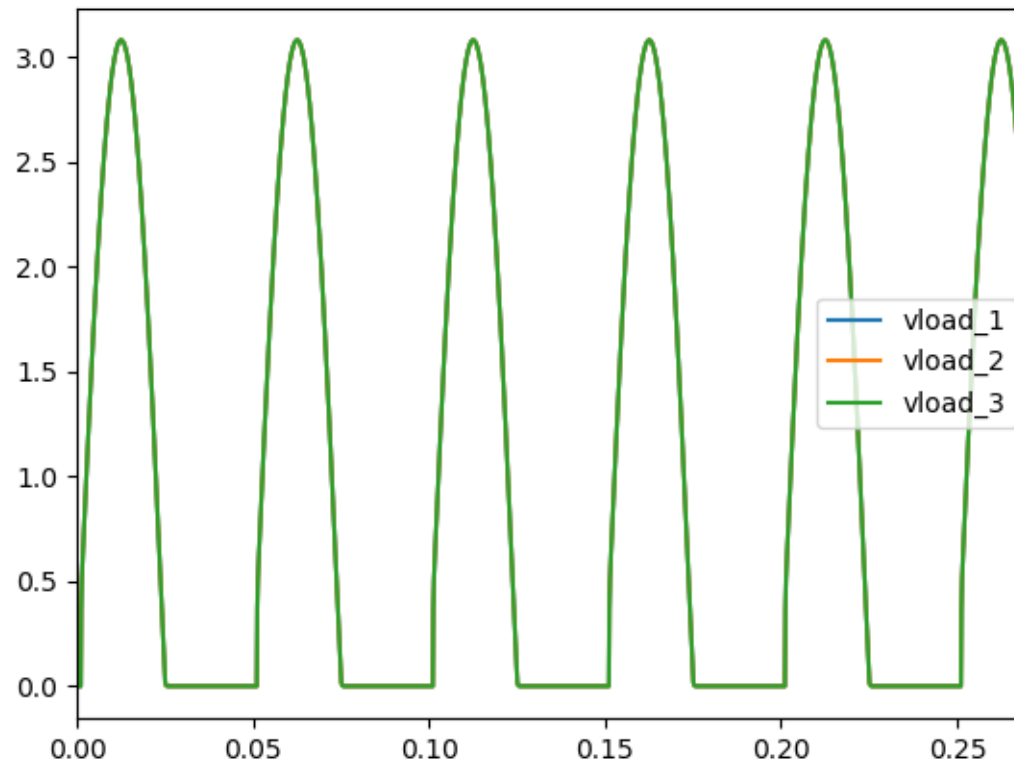
Resistor value = 0,01 Ω (bernilai rendah guna menjaga stabilitas tegangan dan arus)

VoltageSource peak voltage = $V \cdot \sqrt{2}$

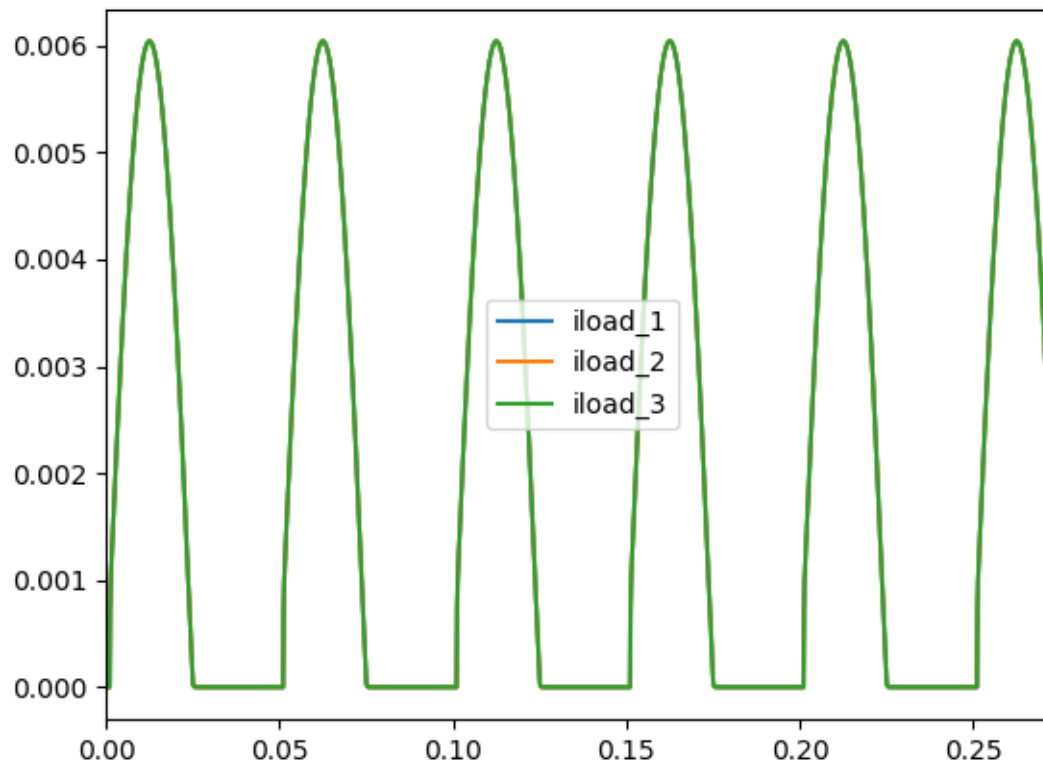
VoltageSource peak voltage = $5 \cdot \sqrt{2}$

VoltageSource peak voltage = 7,071 V

- 3) Simulation output graph
1. Voltage at each resistance



2. The amount of current at each resistance



PPE Simulation Files: <https://github.com/jonathanafernandi/Computational-Physics.git>

Note for Lecturers:

1. The lecturers are advised to assess students' understanding of the topics included in the assignment.
2. The students will submit their answers in .pdf format through BINUSMAYA.
3. The deadline for this comprehensive assignment is at the end of the semester.